

CPR E 4890: Computer Networking and Data Communications

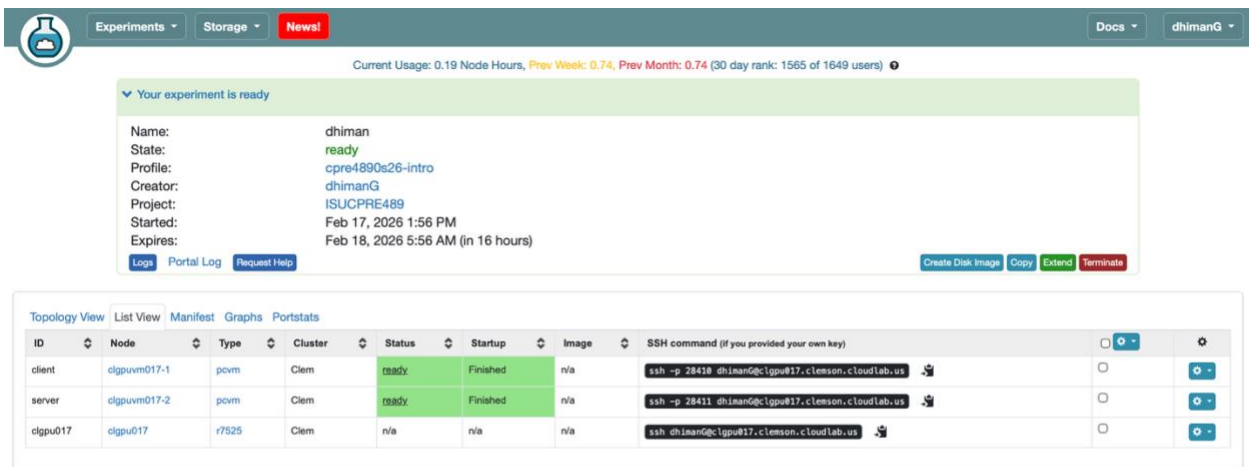
Lab Experiment #4 – Introduction to CloudLab

Summary:

This lab was mainly about CloudLab which is a cloud specific research platform that you can use to create and experiment with virtual network nodes. The main exercises in this lab make us create an experiment with a server and a client node, log into them via SSH (using our own keys), and then running network performance tests using a tool called iperf. The main experiment is to test how bandwidth is impacted when running multiple TCP connections in parallel between a client and server node.

The lab showed me that as you increase the number of parallel TCP connections (from 2 to 10 to 20), the total bandwidth stays roughly constant at around 96 Mb/s, but each individual connection gets a smaller part of that cut. With just 2 connections, each gets around 40-50 Mb/s, but with 20 connections, each connection drops to about 4 Mb/s. The lab also explores network interfaces, which taught me about nodes within CloudLab and how they can communicate with each other, but external machines on the internet cannot directly access CloudLab nodes using their IP addresses.

Exercise 1: Create a Experiment



Current Usage: 0.19 Node Hours, Prev Week: 0.74, Prev Month: 0.74 (30 day rank: 1565 of 1649 users)

Your experiment is ready

Name: dhiman
State: ready
Profile: cpre4890s26-intro
Creator: dhimanG
Project: ISUCPRE489
Started: Feb 17, 2026 1:56 PM
Expires: Feb 18, 2026 5:56 AM (in 16 hours)

Logs Portal Log Request Help

Create Disk Image Copy Extend Terminate

ID	Node	Type	Cluster	Status	Startup	Image	SSH command (if you provided your own key)
client	dlgpuvm017-1	pcvm	Clem	ready	Finished	n/a	ssh -p 28418 dhiman@clgpus017.clemson.ccloudlab.us
server	dlgpuvm017-2	pcvm	Clem	ready	Finished	n/a	ssh -p 28411 dhiman@clgpus017.clemson.ccloudlab.us
dlgpu017	dlgpu017	r7525	Clem	n/a	n/a	n/a	ssh dhiman@clgpus017.clemson.ccloudlab.us

Not sure how to proceed or have further questions? Join the [users group](#) and ask a question. Be sure to include any error messages in your question and the [URL](#) of your experiment status page.

Exercise 2: Logging into your nodes

```
dhiman6@client:~$ ssh -i id_cloudlab_rsa dhiman6@clgpu017.clemson.cloudlab.us -p 28410
Enter passphrase for key 'id_cloudlab_rsa':
Enter passphrase for key 'id_cloudlab_rsa':
Welcome to Ubuntu 22.04.2 LTS (GNU/Linux 5.15.0-160-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

 * Introducing Expanded Security Maintenance for Applications.
   Receive updates to over 25,000 software packages with your
   Ubuntu Pro subscription. Free for personal use.

   https://ubuntu.com/pro
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

dhiman6@client:~$
```

```
dhiman6@server:~$ ssh -i id_cloudlab_rsa dhiman6@clgpu017.clemson.cloudlab.us -p 28411
Enter passphrase for key 'id_cloudlab_rsa':
Welcome to Ubuntu 22.04.2 LTS (GNU/Linux 5.15.0-160-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

 * Introducing Expanded Security Maintenance for Applications.
   Receive updates to over 25,000 software packages with your
   Ubuntu Pro subscription. Free for personal use.

   https://ubuntu.com/pro
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

dhiman6@server:~$
```

Exercise 3: Interact with your nodes

1)

```
dhiman6@client:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.17.17.1 netmask 255.240.0.0 broadcast 172.31.255.255
    inet6 fe80::46:3bff:fe9:9618 prefixlen 64 scopeid 0x20<link>
    ether 02:46:3b:f9:96:18 txqueuelen 1000 (Ethernet)
    RX packets 68143 bytes 49543919 (49.5 MB)
    RX errors 0 dropped 132 overruns 0 frame 0
    TX packets 2157 bytes 186693 (186.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.10.1.1 netmask 255.255.255.0 broadcast 10.10.1.255
    inet6 fe80::8f:d6ff:febb:226e prefixlen 64 scopeid 0x20<link>
    ether 02:8f:d6:bb:22:6e txqueuelen 1000 (Ethernet)
    RX packets 34 bytes 2661 (2.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 11 bytes 1235 (1.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 262 bytes 26244 (26.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 262 bytes 26244 (26.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

dhiman6@server:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.17.17.2 netmask 255.240.0.0 broadcast 172.31.255.255
    inet6 fe80::48:f6ff:fe06:4837 prefixlen 64 scopeid 0x20<link>
    ether 02:48:f6:06:48:37 txqueuelen 1000 (Ethernet)
    RX packets 64009 bytes 49351458 (49.3 MB)
    RX errors 0 dropped 125 overruns 0 frame 0
    TX packets 1861 bytes 167202 (167.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.10.1.2 netmask 255.255.255.0 broadcast 10.10.1.255
    inet6 fe80::86:6cff:fe77:93f2 prefixlen 64 scopeid 0x20<link>
    ether 02:86:6c:77:93:f2 txqueuelen 1000 (Ethernet)
    RX packets 20 bytes 1312 (1.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12 bytes 1325 (1.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 236 bytes 24896 (24.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 236 bytes 24896 (24.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2 and 3) After running `iperf -s` on the server and `iperf -c server -P 2` on the client, we get the output below for both server and client:

```
dhiman6@client:~$ iperf -c server -P 2
[ 1] local 10.10.1.1 port 38494 connected with 10.10.1.2 port 5001
[ 2] local 10.10.1.1 port 38504 connected with 10.10.1.2 port 5001

Client connecting to server, TCP port 5001
TCP window size: 136 KByte (default)

-----
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.0000-0.0689 sec 62.6 MBytes 52.2 Mbits/sec
[ 2] 0.0000-0.0858 sec 52.3 MBytes 43.5 Mbits/sec
[SUM] 0.0000-0.0428 sec 115 MBytes 96.8 Mbits/sec
[ CT] final connect times (min/avg/max/stddev) = 0.716/0.750/0.785/0.049 ms (tot/err) = 2/0

dhiman6@server:~$ iperf -s
Server listening on TCP port 5001
TCP window size: 128 KByte (default)

-----
[ 1] local 10.10.1.2 port 5001 connected with 10.10.1.1 port 38494
[ 2] local 10.10.1.2 port 5001 connected with 10.10.1.1 port 38504
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.0000-0.0565 sec 62.6 MBytes 52.2 Mbits/sec
[ 2] 0.0000-0.0746 sec 52.3 MBytes 43.5 Mbits/sec
[SUM] 0.0000-0.0746 sec 115 MBytes 95.7 Mbits/sec
```

4) After changing the number in the command from 2 TCP connections to 10, we get the following output:

```
dhiman@client:~$ iperf -c server -P 10
[ 2] local 10.10.1.1 port 33922 connected with 10.10.1.2 port 5001
[ 1] local 10.10.1.1 port 33920 connected with 10.10.1.2 port 5001
[ 4] local 10.10.1.1 port 33938 connected with 10.10.1.2 port 5001
[ 7] local 10.10.1.1 port 33958 connected with 10.10.1.2 port 5001
[ 9] local 10.10.1.1 port 33982 connected with 10.10.1.2 port 5001
[10] local 10.10.1.1 port 33996 connected with 10.10.1.2 port 5001
[ 3] local 10.10.1.1 port 33932 connected with 10.10.1.2 port 5001
[ 6] local 10.10.1.1 port 33944 connected with 10.10.1.2 port 5001
-----
Client connecting to server, TCP port 5001
TCP window size: 85.0 KByte (default)
-----
[ 5] local 10.10.1.1 port 33934 connected with 10.10.1.2 port 5001
[ 8] local 10.10.1.1 port 33970 connected with 10.10.1.2 port 5001
[ ID] Interval      Transfer      Bandwidth
[ 6] 0.0000-10.1873 sec 26.6 MBytes  21.9 Mbits/sec
[ 4] 0.0000-10.2196 sec 35.6 MBytes  29.2 Mbits/sec
[ 2] 0.0000-10.2521 sec  7.25 MBytes   5.93 Mbits/sec
[ 5] 0.0000-10.2520 sec 15.5 MBytes  12.7 Mbits/sec
[ 3] 0.0000-10.2682 sec  7.25 MBytes   5.92 Mbits/sec
[10] 0.0000-10.2842 sec  5.00 MBytes   4.08 Mbits/sec
[ 1] 0.0000-10.2845 sec  5.00 MBytes   4.08 Mbits/sec
[ 8] 0.0000-10.2846 sec  5.00 MBytes   4.08 Mbits/sec
[ 7] 0.0000-10.2851 sec  5.00 MBytes   4.08 Mbits/sec
[ 9] 0.0000-10.2847 sec  5.00 MBytes   4.08 Mbits/sec
[SUM] 0.0000-10.1961 sec 117 MBytes  96.5 Mbits/sec
[ CT] final connect times (min/avg/max/stdev) = 0.597/0.871/1.065/0.145 ms (tot/err) = 10/0
```

Also, trying with 20 TCP connections in parallel instead of 10, we get the following output:

```
dhiman@client:~$ iperf -c server -P 20
[ 2] local 10.10.1.1 port 56216 connected with 10.10.1.2 port 5001
[ 3] local 10.10.1.1 port 56236 connected with 10.10.1.2 port 5001
[ 5] local 10.10.1.1 port 56250 connected with 10.10.1.2 port 5001
[ 7] local 10.10.1.1 port 56256 connected with 10.10.1.2 port 5001
[ 9] local 10.10.1.1 port 56270 connected with 10.10.1.2 port 5001
-----
Client connecting to server, TCP port 5001
TCP window size: 85.0 KByte (default)
-----
[12] local 10.10.1.1 port 56278 connected with 10.10.1.2 port 5001
[14] local 10.10.1.1 port 56302 connected with 10.10.1.2 port 5001
[16] local 10.10.1.1 port 56324 connected with 10.10.1.2 port 5001
[17] local 10.10.1.1 port 56338 connected with 10.10.1.2 port 5001
[20] local 10.10.1.1 port 56352 connected with 10.10.1.2 port 5001
[ 1] local 10.10.1.1 port 56228 connected with 10.10.1.2 port 5001
[ 6] local 10.10.1.1 port 56252 connected with 10.10.1.2 port 5001
[11] local 10.10.1.1 port 56274 connected with 10.10.1.2 port 5001
[13] local 10.10.1.1 port 56292 connected with 10.10.1.2 port 5001
[18] local 10.10.1.1 port 56328 connected with 10.10.1.2 port 5001
[19] local 10.10.1.1 port 56346 connected with 10.10.1.2 port 5001
[ 4] local 10.10.1.1 port 56246 connected with 10.10.1.2 port 5001
[10] local 10.10.1.1 port 56276 connected with 10.10.1.2 port 5001
[ 8] local 10.10.1.1 port 56268 connected with 10.10.1.2 port 5001
[15] local 10.10.1.1 port 56308 connected with 10.10.1.2 port 5001
[ ID] Interval      Transfer      Bandwidth
[15] 0.0000-10.0627 sec  7.00 MBytes   5.84 Mbits/sec
[13] 0.0000-10.1432 sec  5.13 MBytes   4.24 Mbits/sec
[16] 0.0000-10.1753 sec  5.38 MBytes   4.43 Mbits/sec
[ 7] 0.0000-10.1923 sec  5.63 MBytes   4.63 Mbits/sec
[14] 0.0000-10.1916 sec  5.88 MBytes   4.84 Mbits/sec
[ 8] 0.0000-10.2083 sec  6.50 MBytes   5.34 Mbits/sec
[12] 0.0000-10.2082 sec  5.50 MBytes   4.52 Mbits/sec
[10] 0.0000-10.2244 sec  6.13 MBytes   5.03 Mbits/sec
[ 1] 0.0000-10.2253 sec  6.13 MBytes   5.02 Mbits/sec
[ 9] 0.0000-10.2252 sec  6.25 MBytes   5.13 Mbits/sec
[18] 0.0000-10.2245 sec  5.00 MBytes   4.10 Mbits/sec
[ 5] 0.0000-10.2410 sec  6.13 MBytes   5.02 Mbits/sec
[19] 0.0000-10.2561 sec  5.38 MBytes   4.40 Mbits/sec
[ 2] 0.0000-10.2894 sec  6.00 MBytes   4.89 Mbits/sec
[ 6] 0.0000-10.3055 sec  6.30 MBytes   5.19 Mbits/sec
[ 3] 0.0000-10.3221 sec  5.13 MBytes   4.17 Mbits/sec
[ 4] 0.0000-10.3219 sec  6.13 MBytes   4.98 Mbits/sec
[20] 0.0000-10.3549 sec  6.13 MBytes   4.96 Mbits/sec
[17] 0.0000-10.3530 sec  5.75 MBytes   4.66 Mbits/sec
[11] 0.0000-10.3541 sec  6.38 MBytes   5.16 Mbits/sec
[SUM] 0.0000-10.2966 sec 118 MBytes  96.0 Mbits/sec
[ CT] final connect times (min/avg/max/stdev) = 0.613/1.075/1.723/0.294 ms (tot/err) = 20/0
```

This tells us that after testing with multiple different # of parallel TCP connections, the average time increases while the total bandwidth stays the same (SUM = 96.0 Mbits/sec), each individual connection is impacted. When we only had 2 parallel connections our bandwidth was around 43.5 Mbits/sec for one connection and 52.2 Mbits/sec for the other. However, when we went up to 20 parallel connections our bandwidth took a much more significant hit and went down to nearly 4 Mbits/sec for nearly all connections that were running in parallel.

5) For each of the three interfaces for the server machine:

i) Identify if this interface can be accessed by the client

```
dhimanG@client:~$ ping 172.17.17.2
PING 172.17.17.2 (172.17.17.2) 56(84) bytes of data.
64 bytes from 172.17.17.2: icmp_seq=1 ttl=64 time=0.959 ms
64 bytes from 172.17.17.2: icmp_seq=2 ttl=64 time=0.305 ms
64 bytes from 172.17.17.2: icmp_seq=3 ttl=64 time=0.323 ms
^C
--- 172.17.17.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2014ms
rtt min/avg/max/mdev = 0.305/0.529/0.959/0.304 ms
dhimanG@client:~$ ping 10.10.1.2
PING 10.10.1.2 (10.10.1.2) 56(84) bytes of data.
64 bytes from 10.10.1.2: icmp_seq=1 ttl=64 time=0.313 ms
64 bytes from 10.10.1.2: icmp_seq=2 ttl=64 time=0.298 ms
64 bytes from 10.10.1.2: icmp_seq=3 ttl=64 time=0.299 ms
^C
--- 10.10.1.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2030ms
rtt min/avg/max/mdev = 0.298/0.303/0.313/0.006 ms
```

As for if the client can access the server, yes. It is able to ping the server. Both IP addresses that are allocated to the Server 172.17.17.2 and 10.10.1.2 are successfully returning our ping requests meaning that the client is able to access our cloudlab server. This is possible because both our cloudlab (client and server) nodes are on the same internal network.

ii) Identify if this interface can be accessed by a machine over the internet

```
> ~ ssh 172.17.17.2
^C
> ~ ping 172.17.17.2
PING 172.17.17.2 (172.17.17.2): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
^C
--- 172.17.17.2 ping statistics ---
4 packets transmitted, 0 packets received, 100.0% packet loss
> ~ ping 10.10.1.2
PING 10.10.1.2 (10.10.1.2): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
^C
--- 10.10.1.2 ping statistics ---
4 packets transmitted, 0 packets received, 100.0% packet loss
> ~
```

However, a machine on the internet, such as my own personal computer (as shown above), is unable to access the client machine created in cloudlab using its IP Addresses. This is also due to the fact that cloudlab nodes are on their own internal IP addresses which are not accessible using the public internet. They belong to cloudlab's internal network.

Exercise 4: Clean up your work

